Min Woo Park

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SHORT BIO. My research focuses on causal decision making (NeurIPS-25, Preprint, ICLR-26) and causal effect identification (NeurIPS-24). I investigate how to formally model diverse real-world problems within a causal framework and derive optimal decisions from such structured representations. In particular, I develop methods for robust decision-making in settings under environmental uncertainty (NeurIPS-25), domain shifts between training and deployment environments (Preprint), and even scenarios involving counterfactual actions (ICLR-26).

EDUCATIONS

Graduate School of Data Science, Seoul National University — Seoul, Republic of Korea
Integrated M.S. and Ph.D. in Data Science — Sep 2022 - Aug 2027 (expected)
Advisor: Prof. Sanghack Lee.
Overall GPA: 4.02 /4.3.

Department of Mathematics and Statistics, Sejong University — Seoul, Republic of Korea
B.S. in Applied statistics and Mathematics (Double major), *Cum Laude* — Mar 2016 - Aug 2022
Advisors: Prof. Ji Eun Lee and Jin Hee Yoon.
Overall GPA: 4.04/4.5; Major GPA: 4.46 (Applied Statistics), 4.38 (Mathematics).

PUBLICATIONS

* for joint first authorship and † for corresponding author.

- ◇ Min Woo Park and Sanghack Lee†. Counterfactual Structural Causal Bandits. *International Conference on Learning Representations* (ICLR), 2026.
- ◇ Min Woo Park, Andy Arditi, Elias Bareinboim†, Sanghack Lee†. Structural Causal Bandits under Markov Equivalence. *Neural Information Processing Systems* (NeurIPS), 2025, Columbia CausalAI Laboratory, Technical Report (R-122)
- ◇ Min Woo Park and Sanghack Lee†. On Transportability for Structural Causal Bandits. *arXiv preprint*, 2025. under review.
- ◇ Yonghan Jung†, Min Woo Park, and Sanghack Lee†. Complete Graphical Criterion for Sequential Covariate Adjustment in Causal Inference. *Neural Information Processing Systems* (NeurIPS), 2024.
- ◇ Min Woo Park*, and Ji Eun Lee*†. Computation of the iterated Aluthge, Duggal, and Mean transforms. Filomat, 2023.
- ◇ Min Woo Park*, Taehui Yun*, Youngin Jang, Younsuk Yeom, Jonghwan Kim, LG AI Research, and Sanghack Lee†. Breaking Bad: Component-wise Parent Deletion for Score-Based Causal Discovery. under review.
- ◇ Yesong Choe, Yeahoon Kwon, Min Woo Park, and Sanghack Lee†. Canonical Domain Reduction for Partial Counterfactual Identification. under review.
- ◇ Ji Eun Lee*† and Min Woo Park*. Convergence of the iterated mean transforms of a 2×2 matrix. under review.

AWARDS AND HONORS

Graduate Student Instructor (GSI) scholarship | Seoul National University — Spring 2024, Spring 2025, Spring 2026

BK21 scholarship — Fall 2024, Fall 2025

First place merit scholarship | Sejong University — Fall 2020, Fall 2021

Fourth place award | The 9th College of Natural Sciences Academic Conference, Sejong University — Nov 2021
▷ Mitigating spurious regression in time-series data via Granger causality.

Third place award | The 8th College of Natural Sciences Academic Conference, Sejong University — Nov 2020
▷ The significance of the Euler's number and its applications.

ACADEMIC SERVICES

Conference Reviewer: 2026 ICLR, ICML, **2025** ICML

PROJECTS

Towards More Scalable Score-based Causal Discovery | LG AI Research — Sep 2025 - Aug 2026

- ▷ Leading the project on developing scalable score-based causal discovery algorithms.
  - Developed a novel operator that enables score-based causal discovery algorithms to escape local optima. (First-author submission under review: "Breaking Bad: Component-wise Parent Deletion for Score-Based Causal Discovery")
  - Investigating parallelization strategies for score-based causal discovery algorithms.

Hyundai NGV Data Boot-Campus | Hyundai Motors — July 2025 - Sep 2025

- ▷ Development of high-voltage battery performance prediction technology.

Hyundai NGV Data Boot-Campus | Hyundai Motors — July 2024 - Sep 2024

- ▷ Development of an intelligent system for predicting vulnerable regions of automotive body panel stiffness through differential geometry-based automatic curvature analysis.

TEACHING ASSISTENTS

Causal Inference for Data Science — Spring 2024, Fall 2024, Spring 2025, Spring 2026

- ▷ Bayesian Networks and Pearl's Graphical Causal Inference Framework.

Machine Learning and Deep Learning for Data Science II — Fall 2023

- ▷ Kernel Methods, Gaussian Processes, and Related Topics for Machine Learning.

Basic Mathematics for Artificial Intelligence — Spring 2022

- ▷ Linear Algebra, Basic Statistics, and Probability Theory for Artificial Intelligence.

PROFESSIONAL TALKS AND POSTERS

Counterfactual Structural Causal Bandits

ICLR 2026 poster session | Rio de Janeiro, Brazil — Apr 2026 (expected)

Structural Causal Bandits under Markov Equivalence

ExploreCSR poster session | Seoul, Republic of Korea — Dec 2025

NeurIPS 2025 poster session | San Diego, USA — Dec 2025

JKAIA 2025, NeurIPS preview session | Seoul, Republic of Korea — Nov 2025

SNU GSDS student research seminar | Seoul, Republic of Korea — Apr 2025

Complete Graphical Criterion for Sequential Covariate Adjustment in Causal Inference

NeurIPS 2024 poster session | Vancouver, Canada — Dec 2024

SNU GSDS student research seminar | Seoul, Republic of Korea — Dec 2024

REFERENCE

Sanghack Lee | Graduate School of Data Science, Seoul National University — sanghack@snu.ac.kr
Associate professor

Ji Eun Lee | Department of Mathematics and Statistics, Sejong University — jieunlee7@sejong.ac.kr
Associate Professor

Jin Hee Yoon | Dept. of AI and Information Technology, Sejong University — jin9135@sejong.ac.kr
Associate Professor

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